

# Lecture 04: Schrödinger's equation & particle in a box

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## Introducing the wave function $\Psi$

- ▶ Electron described by complex wave function  $\Psi(\mathbf{r}, t)$  at every point in space
- ▶  $\Psi$  itself is not directly observable—it is a mathematical construct
- ▶ Contains all information about the quantum state of the electron

### Wave Function

$$\Psi(\mathbf{r}, t) : \mathbb{R}^4 \rightarrow \mathbb{C}$$

#### Notation:

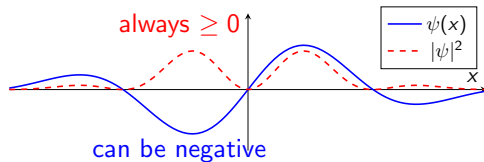
- ▶  $\mathbf{r}$  = position vector  $(x, y, z)$
- ▶  $t$  = time
- ▶  $\mathbb{C}$  = complex numbers (amplitude and phase)

## The probabilistic interpretation: Born rule

- ▶  $|\Psi|^2$  gives probability density of finding electron at position  $\mathbf{r}$
- ▶ Probability in volume  $dV$  equals  $|\Psi|^2 dV$ —always real and non-negative
- ▶ Wave function must be normalized: total probability equals one

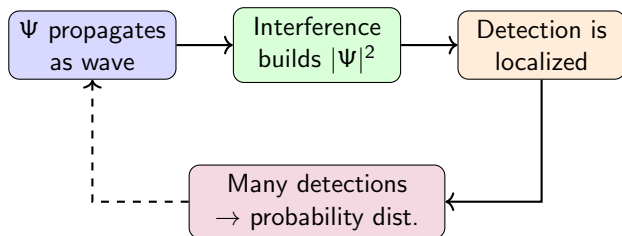
### Born Rule

$$P(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2 = \Psi^* \Psi$$



## Resolving wave-particle duality

- ▶ Wave function propagates as wave—explains interference and diffraction
- ▶ Measurement collapses to particle-like detection at specific location
- ▶ Many measurements reproduce  $|\psi|^2$  distribution: probabilistic predictions



## Checkpoint: wave function interpretation

- ▶ Why is  $|\psi|^2$  observable but  $\psi$  itself is not directly measurable?
- ▶ How does the probabilistic interpretation resolve wave-particle duality?
- ▶ What physical constraint requires the wave function to be normalized?

### Normalization Condition

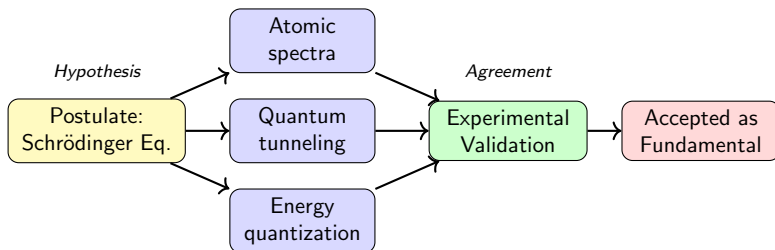
$$\int_{\text{all space}} |\Psi|^2 dV = 1$$

**Key insight:** The electron must be *somewhere*—total probability = 1.

**Think about:** What happens to  $|\psi|^2$  if we multiply  $\psi$  by a phase factor  $e^{i\phi}$ ?

# The Schrödinger equation: a fundamental postulate

- ▶ Schrödinger equation is **NOT** derived from more fundamental principles
- ▶ It is a *postulate*—a brilliant hypothesis validated by experimental agreement
- ▶ Similar to Newton's laws: accepted because predictions match observations



# The time-dependent Schrödinger equation

- ▶ Governs evolution of quantum state  $\Psi(\mathbf{r}, t)$  in space and time
- ▶ Complex-valued wave function: both amplitude and phase carry information
- ▶ First-order in time, second-order in space (unlike classical wave equation)

## Time-Dependent Schrödinger Equation

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

**Left side:** Kinetic + Potential energy acting on  $\Psi$

**Right side:** Time evolution of  $\Psi$

## Notation: key symbols in Schrödinger equation

- ▶  $\hbar = h/2\pi \approx 1.055 \times 10^{-34}$  J·s (reduced Planck constant)
- ▶  $m$  = electron mass (or effective mass in solids)
- ▶  $V = V(\mathbf{r}, t)$  potential energy function (environment)
- ▶  $\Psi = \Psi(\mathbf{r}, t)$  complex wave function;  $|\Psi|^2 dV$  = probability in  $dV$

## The Laplacian Operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

**Physical meaning:**  $-\frac{\hbar^2}{2m}\nabla^2\Psi$  represents kinetic energy operator acting on  $\Psi$

## Separating space and time in Schrödinger equation

- ▶ **Goal:** Find solutions with definite energy from time-dependent equation
- ▶ Assume separable form:  $\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot T(t)$
- ▶ Separation works when potential  $V$  is time-independent

### Separable Ansatz

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot T(t)$$

**Key insight:** Substituting into Schrödinger equation:

$$\underbrace{\frac{1}{\psi} \left[ -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right]}_{\text{depends only on } \mathbf{r}} = \underbrace{\frac{i\hbar}{T} \frac{dT}{dt}}_{\text{depends only on } t} = E$$

Each side must equal the *same constant*  $E$  (the energy eigenvalue)

## Separation of variables: the derivation

- ▶ Substitute  $\Psi = \psi T$  into time-dependent Schrödinger equation
- ▶ Divide both sides by  $\psi T$  to isolate spatial and temporal parts
- ▶ Left side depends only on  $\mathbf{r}$ ; right side depends only on  $t$
- ▶ Both sides must equal separation constant  $E$  (the energy)

**Separated form:**

$$\frac{1}{\psi} \left[ -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right] = \frac{i\hbar}{T} \frac{dT}{dt} = E \quad (1)$$

## Time-independent Schrödinger equation

- ▶ Spatial part yields eigenvalue equation: operator acting on  $\psi$  gives  $E \cdot \psi$
- ▶  $E$  is the energy eigenvalue—a measurable quantity
- ▶ Solutions  $\psi(\mathbf{r})$  are energy eigenfunctions with definite energy  $E$
- ▶ This is the central equation for finding allowed quantum states

## Time-Independent Schrödinger Equation

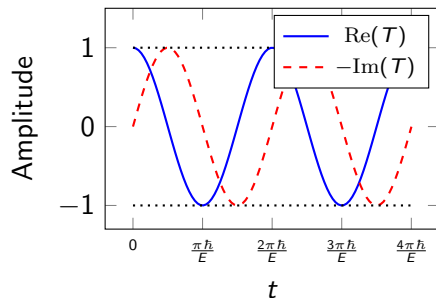
$$\boxed{-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi} \quad (2)$$

### Notation:

- ▶  $\psi(\mathbf{r})$ : spatial wave function
- ▶  $E$ : energy eigenvalue (constant)
- ▶  $V$ : potential energy

## Time evolution: the oscillating phase factor

- ▶ Temporal equation:  $i\hbar \frac{dT}{dt} = ET$  has exponential solution
- ▶ Time dependence is purely oscillatory phase factor
- ▶ Full solution:  $\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot e^{-iEt/\hbar}$
- ▶ Probability density  $|\Psi|^2 = |\psi|^2$  is time-independent!

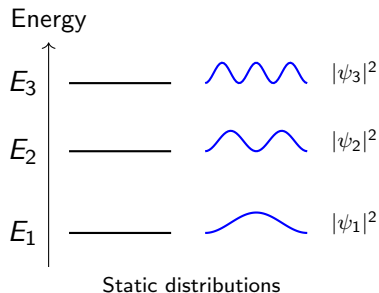


**Time evolution factor:**

$$T(t) = e^{-iEt/\hbar} \quad (3)$$

## Stationary states: why the name?

- ▶ Stationary states have time-independent probability distributions
- ▶ Energy  $E$  is precisely defined—no uncertainty in energy measurement
- ▶ All expectation values  $\langle A \rangle$  are constant in time
- ▶ General solutions: superpositions of stationary states with different  $E$



### Probability density:

$$|\Psi(\mathbf{r}, t)|^2 = |\psi(\mathbf{r})|^2 \quad (4)$$

## Expectation values: quantum averages

**Quantum mechanics predicts average outcomes** over many measurements.

### Expectation Value Definition

$$\langle A \rangle = \frac{\int_V \psi^* \hat{A} \psi dV}{\int_V |\psi|^2 dV} \quad (5)$$

#### Notation:

- ▶  $\hat{A}$  = quantum operator for observable  $A$
- ▶ For normalized  $\psi$ : denominator = 1

#### Common examples:

$$\langle z \rangle = \int \psi^* z \psi dV \quad (\text{average position}) \quad (6)$$

$$\langle p \rangle = \int \psi^* (-i\hbar \nabla) \psi dV \quad (\text{average momentum}) \quad (7)$$

## Pause

Given the Schrödinger's equation, now we can compute a few examples on what happens with the electron under multiple assumptions. In particular, we will look at what happens if we place the electrons in different potentials

## Free electron wave function ( $V = 0$ )

- ▶ **Single region (all  $z$ ):**  $V(x) = 0 \Rightarrow$  no interface, no scattering, no reflection
- ▶ **Plane-wave solution:**

$$\psi(x, t) = A e^{i(kx - \omega t)}$$

- ▶ **Stationary-state form (spatial part):**

$$\psi(z) = A e^{ikx}$$

- ▶ General free-particle solution is a superposition of right- and left-moving waves,

$$\psi(z) = A e^{ikx} + B e^{-ikx},$$

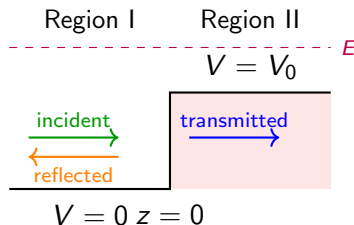
but for a *single* plane wave traveling to the right we take  $B = 0$ .

## Dispersion and wave vector

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m} \quad (8)$$

## Potential step: setup and classical expectation

- ▶ Electron with energy  $E$  approaches potential step  $V_0$  at  $z = 0$
- ▶ **Classical:** full transmission if  $E > V_0$ , full reflection if  $E < V_0$
- ▶ **Quantum:** wave nature creates partial reflection even when  $E > V_0$
- ▶ Must solve Schrödinger equation in each region and match at boundary



## Wave functions in each region ( $E > V_0$ )

- ▶ **Region I** ( $z < 0$ ):  $\psi_1 = Ae^{ik_1z} + Be^{-ik_1z}$  with  $k_1^2 = 2mE/\hbar^2$
- ▶ **Region II** ( $z > 0$ ):  $\psi_2 = Ce^{ik_2z}$  with  $k_2^2 = 2m(E - V_0)/\hbar^2$
- ▶ No leftward wave in Region II (no source at  $z = +\infty$ )
- ▶  $A$  = incident amplitude,  $B$  = reflected,  $C$  = transmitted

## Wave Vectors in Each Region

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}}, \quad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \quad (9)$$

**Note:**  $k_2 < k_1$  when  $E > V_0$  (slower wave in Region II)

## Boundary conditions at $z = 0$

- ▶ **Continuity of  $\psi$ :**  $\psi_1(0) = \psi_2(0) \rightarrow A + B = C$
- ▶ **Continuity of  $d\psi/dz$ :** ensures probability current conservation
- ▶ At  $z = 0$ :  $ik_1(A - B) = ik_2C$
- ▶ Two equations, three unknowns  $\rightarrow$  express  $B$  and  $C$  in terms of  $A$

## Boundary Conditions at Interface

$$A + B = C \quad (10)$$

$$k_1(A - B) = k_2C \quad (11)$$

**Physical meaning:** Wave function and its slope must be continuous for finite potential steps.

## Reflection and transmission coefficients

- ▶ Solve boundary equations:  $\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}$ ,  $\frac{C}{A} = \frac{2k_1}{k_1 + k_2}$
- ▶ Reflection coefficient  $R = |B/A|^2$  measures reflected probability
- ▶ Transmission coefficient  $T = (k_2/k_1)|C/A|^2$  accounts for velocity change
- ▶ Verify  $R + T = 1$ : probability conservation satisfied

### Reflection and Transmission Coefficients

$$R = \left( \frac{k_1 - k_2}{k_1 + k_2} \right)^2, \quad T = \frac{4k_1 k_2}{(k_1 + k_2)^2} \quad (12)$$

**Check:**  $R + T = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2} = \frac{(k_1 + k_2)^2}{(k_1 + k_2)^2} = 1 \quad \checkmark$

## Checkpoint: boundary condition application

### Conceptual Questions

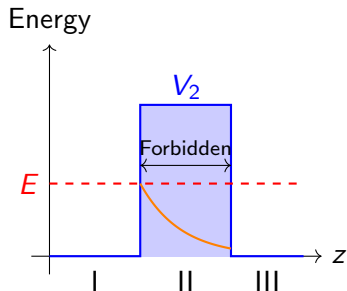
1. Why must both  $\psi$  and  $d\psi/dz$  be continuous at the boundary?
2. What happens to  $R$  as  $E$  approaches  $V_0$  from above?
3. Why is there no  $D \cdot e^{-ik_2z}$  term in Region II?
4. How does this result connect to optical reflection at interfaces?

**Hint for Q4:** Compare  $R = \left( \frac{k_1 - k_2}{k_1 + k_2} \right)^2$  with Fresnel equations for normal incidence:

$$R = \left( \frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

## Classically forbidden regions: when $E < V$

- ▶ Classical particle cannot exist where kinetic energy would be negative ( $E < V$ )
- ▶ Quantum mechanics: wave function doesn't vanish—it **decays exponentially**
- ▶ Region where  $E < V_2$  is called the **classically forbidden region**
- ▶ Probability density  $|\psi|^2$  remains finite inside barrier



## Wave function inside the barrier

**Schrödinger equation inside barrier:**

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} + V_2\psi = E\psi \quad (13)$$

**Rearranging:**

$$\frac{d^2\psi}{dz^2} = \frac{2m}{\hbar^2}(V_2 - E)\psi = \gamma^2\psi \quad \text{where } \gamma^2 > 0 \quad (14)$$

**Decay Constant in Barrier**

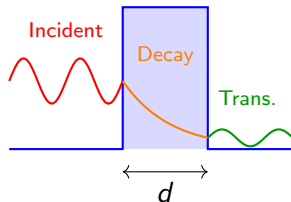
$$\gamma = \sqrt{\frac{2m(V_2 - E)}{\hbar^2}} \quad (15)$$

**Solution:** Real exponentials (not oscillatory)

$$\psi_2 = Ae^{-\gamma z} + Be^{+\gamma z} \quad (16)$$

# Exponential decay through finite barrier

- ▶ For barrier of width  $d$ : keep both exponential terms inside
- ▶ Wave function decays as  $e^{-\gamma z}$  through barrier
- ▶ Amplitude at exit reduced by factor  $e^{-\gamma d}$
- ▶ Probability reduced by  $e^{-2\gamma d}$



## Amplitude Decay

$$\psi_2(z) \propto e^{-\gamma z}, \quad |\psi|^2 \propto e^{-2\gamma d}$$

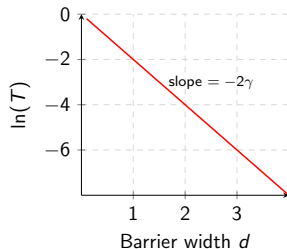
# Tunneling transmission probability

- ▶ Apply boundary conditions at both interfaces ( $z = 0$  and  $z = d$ )
- ▶ Transmission coefficient:  $T = \frac{|\text{transmitted}|^2}{|\text{incident}|^2}$
- ▶ For thick barriers ( $\gamma d \gg 1$ ):  $T \approx T_0 e^{-2\gamma d}$

## Tunneling Probability (Thick Barrier)

$$T \approx T_0 \exp\left(-\frac{2d}{\hbar} \sqrt{2m(V_2 - E)}\right)$$

*Doubling barrier width squares the exponential suppression*



# Physical interpretation: energy-time uncertainty

## Energy-Time Uncertainty Relation

$$\Delta E \cdot \Delta t \geq \hbar \quad (17)$$

- ▶ Tunneling doesn't violate energy conservation—uses uncertainty principle
- ▶ Electron “borrows” energy  $\Delta E = V_2 - E$  for time  $\Delta t \approx d/v$
- ▶ Faster traversal (thinner barrier) allows larger energy borrowing
- ▶ Heavier particles tunnel less:  $\gamma \propto \sqrt{m}$  means exponentially suppressed

**Key insight:** Quantum mechanics allows temporary energy fluctuations within the bounds set by the uncertainty principle

## Calculating uncertainties from expectation values

**Uncertainty** = root-mean-square deviation from average:

Uncertainty Formula

$$\Delta z = \sqrt{\langle z^2 \rangle - \langle z \rangle^2} \quad (18)$$

**Procedure:**

1. Calculate  $\langle z \rangle = \int \psi^* z \psi dV$
2. Calculate  $\langle z^2 \rangle = \int \psi^* z^2 \psi dV$
3. Compute  $\Delta z = \sqrt{\langle z^2 \rangle - \langle z \rangle^2}$

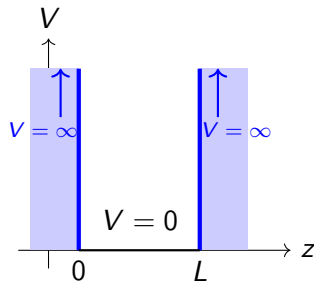
**Same for momentum:**

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} \quad (19)$$

Product  $\Delta z \Delta p \geq \hbar$  always satisfied for any valid  $\psi$ .

## Infinite potential well: problem setup

- ▶ Electron confined between rigid walls at  $z = 0$  and  $z = L$  with  $V = \infty$  outside
- ▶ Inside well ( $0 < z < L$ ):  $V = 0$ , electron moves freely
- ▶ **Boundary conditions:**  $\psi$  must vanish at walls
- ▶ Simplest exactly solvable quantum confinement problem



## Solving inside the infinite well

**Time-independent Schrödinger equation with  $V = 0$ :**

Wave Equation Inside Well

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} = E\psi \quad (20)$$

**Rearranging:**

$$\frac{d^2\psi}{dz^2} = -\frac{2mE}{\hbar^2}\psi = -k^2\psi \quad (21)$$

**General solution:**

$$\psi(z) = A \sin(kz) + B \cos(kz) \quad (22)$$

**Wave vector related to energy:**

$$k^2 = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{\hbar^2 k^2}{2m} \quad (23)$$

## Applying boundary conditions

**At  $z = 0$ :**  $\psi(0) = 0$

$$A \sin(0) + B \cos(0) = B = 0 \quad \Rightarrow \quad \psi(z) = A \sin(kz) \quad (24)$$

**At  $z = L$ :**  $\psi(L) = 0$

$$A \sin(kL) = 0 \quad \Rightarrow \quad kL = n\pi \quad (25)$$

### Quantization Condition

$$k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (26)$$

- ▶ **Quantization emerges:** only discrete  $k$  values allowed
- ▶ Quantum number  $n = 1, 2, 3, \dots$  (note:  $n = 0$  gives trivial  $\psi = 0$ )

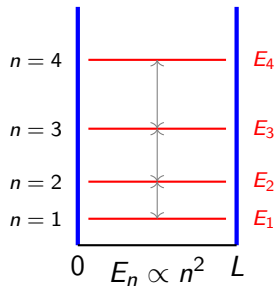
# Quantized energy levels in infinite well

Substituting  $k = n\pi/L$  into  $E = \hbar^2 k^2 / 2m$ :

## Energy Eigenvalues

$$E_n = \frac{h^2 n^2}{8mL^2} = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

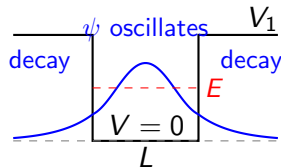
- ▶ Ground state ( $n = 1$ ) has non-zero energy:  
**zero-point energy**
- ▶ Energy spacing increases with  $n$
- ▶ Narrower wells  $\Rightarrow$  higher energies



# Finite potential well: realistic confinement

## Key Features:

- ▶ Finite barrier height  $V_1$  allows wave function penetration into barriers
- ▶ Inside well: oscillatory solution with  $k^2 = \frac{2mE}{\hbar^2}$
- ▶ Outside well: exponential decay with  $\gamma^2 = \frac{2m(V_1 - E)}{\hbar^2}$
- ▶ Bound states exist only for  $E < V_1$ ; **finite number** of discrete levels



## Transcendental equation and graphical solution

**Matching boundary conditions** ( $\psi$  and  $d\psi/dz$  continuous) yields:

$$k \tan \left( \frac{kL}{2} \right) = \gamma \quad (27)$$

where  $k = \sqrt{2mE}/\hbar$  and  $\gamma = \sqrt{2m(V_1 - E)}/\hbar$ .

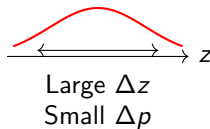
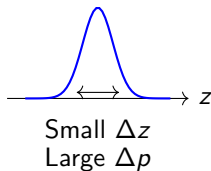
- ▶ No closed-form solution
- ▶ Graphical: plot both sides vs  $E$
- ▶ Intersections  $\rightarrow$  allowed energies
- ▶ Deeper/wider wells  $\rightarrow$  more states

# Reminder: Heisenberg uncertainty principle

## Fundamental Uncertainty Relations

$$\Delta z \Delta p \geq \hbar, \quad \Delta E \Delta t \geq \hbar \quad (28)$$

- ▶ **Position-momentum:** Cannot know both precisely simultaneously
- ▶ **Energy-time:** Short-lived states have uncertain energy
- ▶ Not a measurement limitation—**fundamental property of wave nature**



# Uncertainty principle explains tunneling

## Energy-time uncertainty allows “borrowing”:

- ▶ Temporary energy fluctuation:  
 $\Delta E \sim \hbar / \Delta t$
- ▶ Electron can surmount barrier if transit time is short
- ▶ Thinner barrier  $\rightarrow$  shorter  $\Delta t \rightarrow$  larger  $\Delta E$
- ▶ **Energy conserved on average**—no thermodynamic violation

